



Are deleted files still on your hard disk?

of damage, it can be very expensive as well. Whether your data has been accidentally deleted, your drive been formatted, your entire data swept away by a virus or even the drive no longer being recognised by the operating system – you still don't need to enter into panic mode com-

pletely. Many times, the data is recoverable, and fortunately, there's a bunch of file recovery programs to help you do this on the cheap, if not for free.

How does recovery software work?

So, how does this kind of software work exactly? When you deleted your data from the recycle bin, the command prompt did say that you were deleting it “permanently”. Then how are we able to magically resurrect this piece of information from its grave? Well, the truth is, deleting data is a bit more complicated than simply erasing a piece of information from the drive. Surprisingly, in fact, the data isn't actually removed from the drive when deleted. The files are just not accessible in the way they were anymore. For example, if you delete a file, the file's data isn't erased from the drive, but a signature byte is placed at the location of the file informing the OS that the area can be overwritten by other data next. In other words, the file system marks the areas that previously contained the file as free space that can now be overwritten with new data if need be. The file is now invisible to the operating system and as a result, to you. It's precisely this hidden state of which the file recovery system takes advantage – it knows that though the directions to the file are missing, the file essentially isn't. And this isn't just the case with your computer hard disk. Files are deleted and stored in a similar manner when it comes to your SSDs, external hard drives, iPods, CDs, USBs, etc. So yes, deleted data can be recovered from any of these data storage devices – although, the particular program must support the storage device in question. Recovery works even in what seems like the most extreme of cases – when you've formatted a drive from FAT32 to NTFS, and suddenly realise that you didn't need to do it since all your files were still there. Luckily, when you format a FAT32 drive, everything is destroyed but the partition table and the data – and that's all we need! Since the data's still there, your files can still be reconstructed.